

Intermediate Macroeconomics

BSc. in Business Administration and BSc. in Finance and Accounting

Intermediate Macroeconomics – BMAK

Chapter 4: The Macroeconomy in the Short Run

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The Meaning of Money

In a world without money: Barter economy. Exchanging one good or service for another. Trade requires a double coincidence of wants. Unlikely occurrence that two people each have a good or service that the other wants

Money makes trade easier

Money is a set of assets in an economy that people regularly use to buy goods and services from other people.

The role of expectation: People exchange goods and services for money because they expect that later the money could be exchanged for goods and services they need.

Liquidity: Ease with which an asset can be converted into the economy's medium of exchange

The functions and kinds of Money

Functions of money

1. Medium of exchange: Item that buyers give to sellers when they want to purchase goods and services
2. Unit of account: Yardstick people use to post prices and record debts
3. Store of value: Item that people can use to transfer purchasing power from the present to the future

Kinds of money:

Commodity money: Money that takes the form of a commodity with intrinsic value: gold, cigarettes. Commodity money has intrinsic value, i.e., item would have value even if it were not used as money

Gold standard - Gold as money or paper money that is convertible into gold on demand

Fiat money: Money without intrinsic value. Used as money because of government decree. "This note is legal tender for all debts, public and private".

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old/Money in the economy

Currency: Paper bills and coins in the hands of the public

Demand deposits: Balances in bank accounts; depositors can access on demand by writing a check

Money stock: Quantity of money circulating in the economy

Measures of money stock:

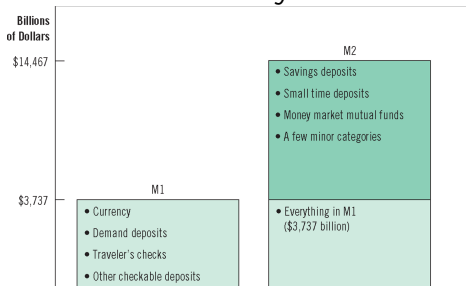


FIGURE 1

Two Measures of the Money Stock for the U.S. Economy

The two most widely followed measures of the money stock are M1 and M2. This figure shows the size of each measure in January 2019.

Source: Federal Reserve.

The two-tier banking system

Central bank: Institution designed to oversee the banking system.

Regulate the quantity of money in the economy The Federal Reserve (the Fed) - the central bank of the United States was created in 1913 after a series of bank failures in 1907 to ensure the health of the nation's banking system.

Commercial banks-financial intermediaries: Can influence the quantity of demand deposits in the economy (and the money supply)

Central bank: Board of governors include 7 members, 14-year terms.

Appointed by the president and confirmed by the Senate. The chairman: Jerome Powell

- Directs the Fed staff

- Presides over board meetings

- Testifies regularly about Fed policy in front of congressional committees.

- Appointed by the president (4-year term)

Central bank in Vietnam

The State Bank of Vietnam

Headed by the governor, Madam Nguyen Thi Hong. And deputy governors.

There are 63 branches in 63 provinces.

The screenshot shows the official website of the State Bank of Vietnam (NHNN). The top navigation bar includes links for News, Policies, Accounting & Treasury, Issuance, Foreign Exchange & Gold Business, and Statistics. The main content area features a congratulatory message from the Governor, titled "Thư chúc mừng của Thống đốc Ngân hàng Nhà nước Việt Nam nhân dịp kỷ niệm 75 năm thành lập Ngân hàng Việt Nam (06/5/1951 - 06/5/2026)". Below the text is a video player showing a portrait of the Governor. To the right, a sidebar lists various services and information, each with a plus icon to expand it.

Thư chúc mừng của Thống đốc Ngân hàng Nhà nước Việt Nam nhân dịp kỷ niệm 75 năm thành lập Ngân hàng Việt Nam (06/5/1951 - 06/5/2026)

29/04/2025 18:24:00

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NGÂN HÀNG VIỆT NAM
75

Thư chúc mừng
CỦA THỐNG ĐỐC NGÂN HÀNG NHÀ NƯỚC
NHÂN DỊP KỶ NIỆM 75 NĂM THÀNH LẬP
NGÂN HÀNG VIỆT NAM (06/5/1951 - 06/5/2026)

Giới thiệu NHNN +

CPI

Lãi suất

Dự trữ bắt buộc

Hoạt động thị trường tiền tệ +

Cải cách hành chính +

Dịch vụ công trực tuyến

Hoạt động khác +

Cơ chế chính sách hỗ trợ +
doanh nghiệp

NHNN với Quốc hội và cử tri +

Công bố thông tin của Doanh +
nghiệp nhà nước thuộc ngành
Ngân hàng

Hệ thống các TCTD +

Chương trình, chính sách tín +
dụng trọng điểm

Phổ biến kiến thức +

Tiếp cận thông tin +

Central bank's job

Regulate banks and ensure the health of the banking system. Monitors each bank's financial condition. Facilitates bank transactions - clearing checks. Acts as a bank's bank - lender of last resort.

Control the money supply, i.e., the quantity of money available in the economy

Conduct monetary policy: by the Federal Open Market Committee (FOMC).
In Vietnam: Monetary Policy Council.

Monetary policy: Setting the interest rate.

central banks' primary tool: open-market operation: Purchase & sale of U.S. government bonds. To increase the money supply: open-market purchase. To decrease the money supply: open-market sale.

The commercial banks and money supply

Defining reserves: Deposits that banks have received but have not loaned out

The simple case of 100% reserve banking: all deposits are held as reserves.

Banks do not influence the supply of money

First National Bank

Assets		Liabilities	
Reserves	\$100.00	Deposits	\$100.00

The actual case of Fractional-reserve banking: Banks hold only a fraction of deposits as reserves

Reserve ratio: Fraction of deposits that banks hold as reserves

Reserve requirement: Minimum amount of reserves that banks must hold; set by the central bank.

The commercial banks and money supply

First National Bank

Assets		Liabilities	
Reserves	\$10.00	Deposits	\$100.00
Loans	90.00		

Example: 10% reserve ratio.

Second National Bank

Assets		Liabilities	
Reserves	\$ 9.00	Deposits	\$90.00
Loans	81.00		

Third National Bank

Assets		Liabilities	
Reserves	\$ 8.10	Deposits	\$81.00
Loans	72.90		

The commercial banks and money supply

The money multiplier: Original deposit = \$100.00. First National lending = \$ 90.00 [= $.9 \times \$100.00$]. Second National lending = \$ 81.00 [= $.9 \times \$90.00$]. Third National lending = \$ 72.90 [= $.9 \times \$81.00$] ... Total money supply = \$1,000.00

The money multiplier: Amount of money the banking system generates with each dollar of reserves. It is the reciprocal of the reserve ratio = $1/R$. The higher the reserve ratio. The smaller the money multiplier.

More Realistic National Bank

Assets		Liabilities and Owners' Equity	
Reserves	\$200	Deposits	\$800
Loans	700	Debt	150
Securities	100	Capital (owners' equity)	50

Leverage and the 2008-2009 Financial Crisis

Leverage: Use of borrowed money to supplement existing funds for purposes of investment

Leverage ratio: Ratio of assets to bank capital.

Capital requirement: Government regulation specifying a minimum amount of bank capital

If bank's assets rise in value by 5% because some of the securities the bank was holding rose in price

\$1,000 of assets would now be worth \$1,050. Bank capital rises from \$50 to \$100

So, for a leverage rate of 20, a 5% increase in the value of assets increases the owners' equity by 100%.

If bank's assets are reduced in value by 5% because some people who borrowed from the bank default on their loans

\$1,000 of assets would be worth \$950. The value of the owners' equity falls to zero.

So, for a leverage ratio of 20, a 5 % fall in the value of the bank assets leads to a 100% fall in bank capital.

Leverage and the 2008-2009 Financial Crisis

If bank's assets are reduced in value by more than 5% because some people who borrowed from the bank default on their loans. For a leverage ratio of 20, the bank's assets would fall below its liabilities. The bank would be insolvent: unable to pay off its debt holders and depositors in full.

Many banks in 2008 and 2009 incurred sizable losses on some of their assets. Mortgage loans and securities backed by mortgage loans drops in value due to the sub-prime lending practice. Shortage of capital induced the banks to reduce lending $\Rightarrow$ Credit crunch: Banks had money but did not dare to lend because they are doubtful of their customers' solvency. $\Rightarrow$ Contributed to a severe downturn in economic activity.

U.S. Treasury and the Fed: Put many billions of dollars of public funds into the banking system to increase the amount of bank capital. Temporarily made the U.S. taxpayer a part owner of many banks. Goal: to recapitalize the banking system so that bank lending could return to a more normal level - occurred by late 2009

Central banks: Monetary Policy Tools

CB: influences the quantity of reserves: Open-market operations or lending to banks. Influences the reserve ratio: Reserve requirements. Paying interest on reserves.

Open-market operations: purchase and sale of government bonds by the central bank. To increase the money supply: the CB buys government bonds. to reduce the money supply: the Fed sells U.S. government bonds. OMOs are easy to conduct and used more often.

CB lending to banks: to increase the money supply via a discount window at the discount rate. Example: Term Auction Facility: to the highest bidder
The discount rate: interest rate on the loans that the Fed makes to banks.
Higher discount rate: reduce the money supply. Smaller discount rate: increase the money supply .

Central banks: Monetary Policy Tools

Example: Term Auction Facility (2007 to 2010) by Fed: The Fed sets a quantity of funds it wants to lend to banks. Eligible banks bid to borrow those funds. Loans go to the highest eligible bidders: acceptable collateral and pay the highest interest rate.

Reserve requirements: the minimum amount of reserves that banks must hold against deposits. An increase in reserve requirement: decrease the money supply. A decrease in reserve requirement: increase the money supply. Used rarely – disrupt the business of banking. This measure is also less effective in recent years because many banks hold excess reserves.

Paying interest on reserves: The higher the interest rate on reserves $\Rightarrow$ the more reserves banks choose to hold. An increase in the interest rate on reserves $\Rightarrow$ increase the reserve ratio $\Rightarrow$ lower the money multiplier $\Rightarrow$ lower the money supply.

Challenges facing CB: CBs control of the money supply is not precise. CB does not control: the amount of money households choose to hold as bank deposits, and the amount that bankers choose to lend.

Modern CBs: The conduct of MP

Conduct monetary policies based on the price, not the quantity.

The federal funds rate: interest rate at which banks make overnight loans to one another. Lender – has excess reserves. Borrower – needs reserves. A change in federal funds rate $\Rightarrow$ Changes other interest rates.

ON rate differs from the discount rate: Affects other interest rates as well. It is determined by supply and demand in the market for loans among banks. Targeted by the CB: Change the ON $\Rightarrow$ change the money supply.

The Fed targets the federal funds rate through open-market operations. CB cuts the ON rate: buy bonds to increase money supply to supply the shortage of money. CB raises ON rates: sells bonds to withdraw the excessive money.

Quantity theory of money

The quantity of money available in the economy determines (the value of money) the price level. Growth rate in quantity of money available determines the inflation rate

Classical Dichotomy: *Nominal variables*: variables measured in monetary units. *Real variables*: variables measured in physical units: relative prices, real wages, real interest rate. *Classical dichotomy*: Theoretical separation of nominal and real variables.

Monetary neutrality: Developments in the monetary system influence nominal variables but are irrelevant for explaining real variables. Changes in money supply don't affect real variables.

Not completely realistic in the short-run

Correct in the long run

Quantity theory of money

New concept: Velocity of money (V) is the rate at which money changes hands.

$$V \times M = (P \times Y)$$

P = price level (GDP deflator)

Y = real GDP

M = quantity of money

An increase in the quantity of money must be reflected in:

- Price level must rise

- Quantity of output must rise

- Velocity of money must fall

Velocity of money: Relatively stable over time

Changes in the quantity of money, M must be proportionate to changes in nominal value of output ($P \times Y$).

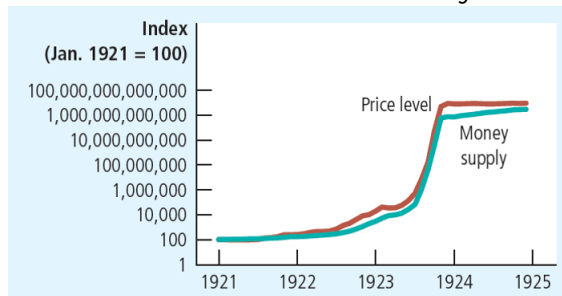
Economy's output of goods & services, Y , primarily determined by factor supplies and available production technology

Quantity theory of money

Money does not affect output

Change in money supply, M , induces proportional changes in the nominal value of output ($P \times Y$). Which is reflected in changes in the price level (P).

When the central bank increases the money supply rapidly $\Rightarrow$ High rate of inflation as the case of Germany after the first world war.



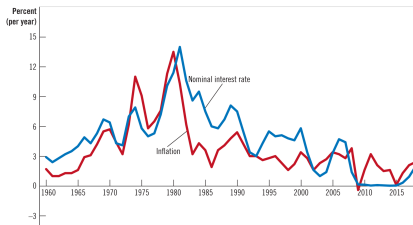
The Fisher Effect

This figure uses annual data since 1960 to show the nominal interest rate on three-month Treasury bills and the inflation rate as measured by the consumer price index. The close association between these two variables provides evidence for the Fisher effect: When the inflation rate rises, so does the nominal interest rate.

Source: U.S. Department of Treasury, U.S. Department of Labor.

FIGURE 5

The Nominal Interest Rate and the Inflation Rate



Principle of monetary neutrality: An increase in the rate of money growth raises the rate of inflation but does not affect any real variable.

Real interest rate = Nominal interest rate – Inflation rate

Nominal interest rate = Real interest rate + Inflation rate

The Financial Sector and the TR-Curve

To pin down short-run macroeconomic outcomes, we complement our analysis of the real sector of the economy, as reflected in the *IS*-curve, with an analysis of the financial sector. For the financial sector, we analyze the roles of the central bank and commercial banks in determining the rate at which households and firms can borrow. Their decision making gives rise to an additional relationship, beyond the one depicted by the *IS*-curve, between output and the interest rate: the *TR*-curve.

We begin by considering central banks. Modern **central banks** typically pursue the **objective** of keeping **inflation low and stable**, while also contributing to an **adequate level of output**. While the latter target is readily appreciated, what are the **benefits of low and stable inflation**?

- + When inflation is high, resources are not allocated efficiently:
 - periods of high inflation are empirically also periods of high price volatility;
 - under high price volatility, it becomes more difficult to identify industries whose prices are falling in relative terms, which is an indicator of high productivity growth. This leads to a less efficient allocation of resources.
- + Inflation devalues money holdings, creating an “inflation tax”. Attempts by households and firms to avoid this tax generate inefficiencies, often referred to as “shoe-leather costs”, as agents reallocate between interest-bearing and non-interest-bearing assets.

Why, then, do central banks aim for a positive rate of inflation?

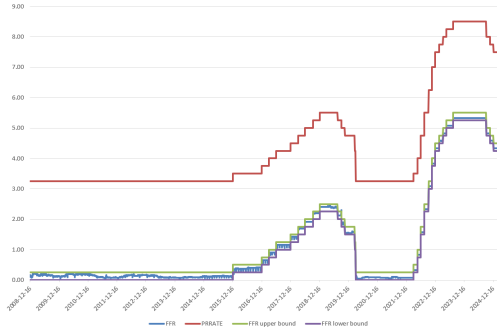
- + With positive inflation, there is a lower risk of deflation. Deflation is costly partly because, for a given level of nominal debt, the real value of debt increases.
- + Labor markets appear to function better under positive inflation, as employees tend to dislike real wage cuts caused by inflation less than explicit nominal wage cuts.

Interest Rate Policy and Inflation Control

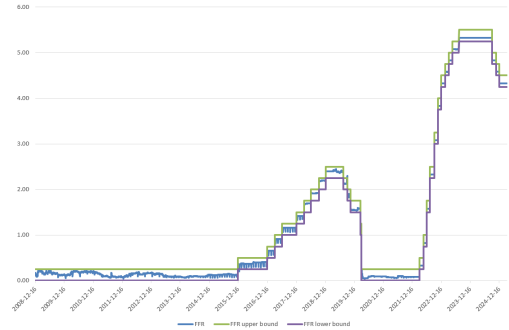
To control inflation, central banks (in "normal times") vary the interest rate at which they are willing to lend to commercial banks. This is based on the following rationale:

- + If the central bank is worried that inflation is above (below) the target rate of inflation, it will raise (lower) the interest rate at which it is willing to lend to commercial banks.
- + The more (less) costly it is for commercial banks to secure funds from the central bank, the higher (lower) the interest rate they charge for loans to households and firms.
- + The higher (lower) the cost of borrowing for households and firms, the less (more) they can fund consumption and investment expenditure through borrowing.
- + Aggregate output demand then will fall (increase), leading to cuts (rises) in the production of output, and eventually also to lower (higher) real wages and thus lower (higher) costs of production, which in turn leads to cuts (increases) in output prices, and thus the desired control of inflation.

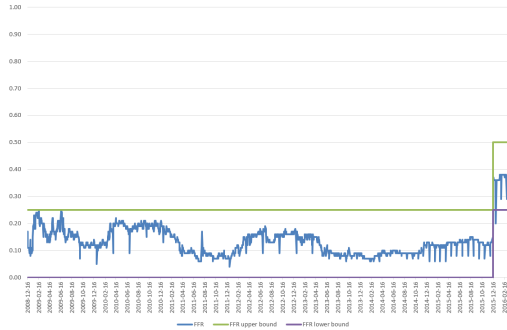
Monetary vs. Lending rate



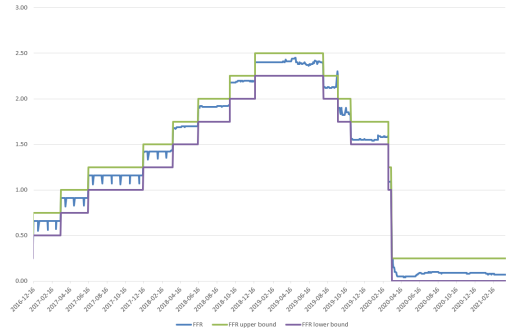
Monetary vs. Lending rate



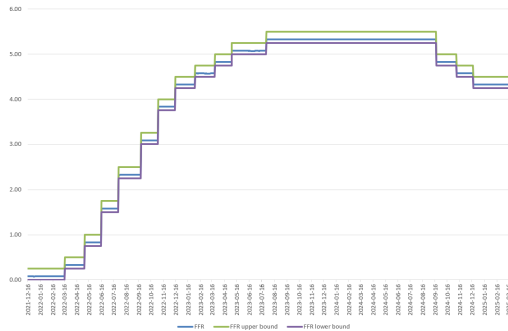
Monetary vs. Lending rate



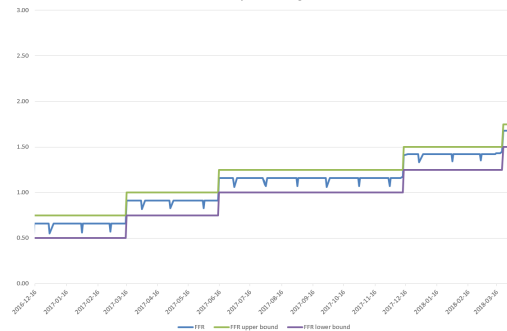
Monetary vs. Lending rate



Monetary vs. Lending rate



Monetary vs. Lending rate



The ECB

To provide further institutional context, let us consider the **European Central Bank's** (ECB's) interest rate operations in normal times:

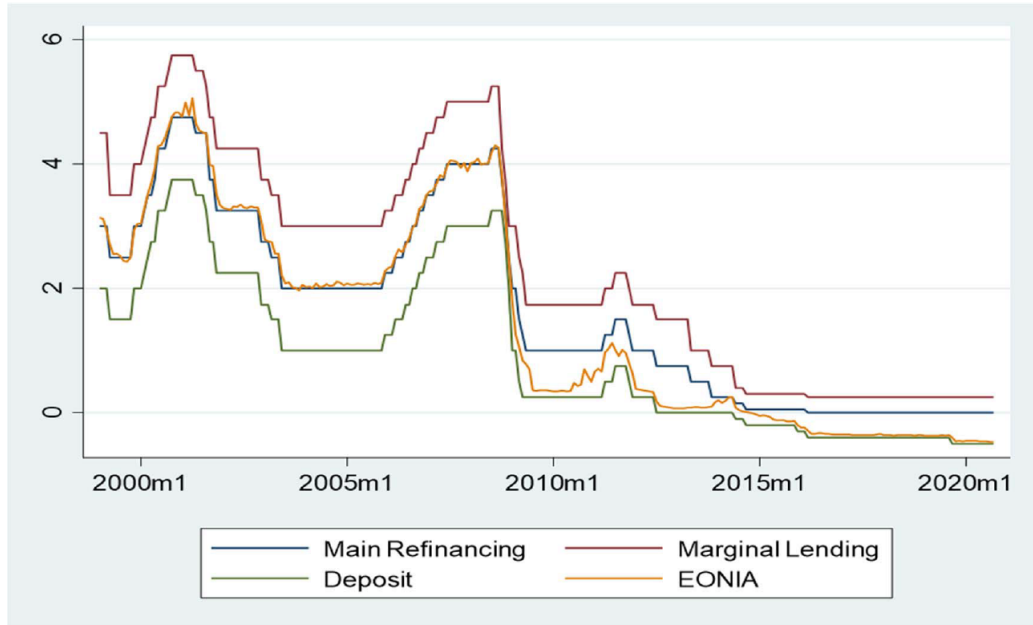
Open Market Operations

- + *Main refinancing operations*: weekly auctions for loans (maturity: one week). Commercial banks submit bids, and the ECB chooses the volume of funds to allocate and at which interest rate.
- + *Longer-term refinancing operations*: monthly auctions for loans (maturity: three months), to complement the main refinancing operations.

Standing Facilities:

- + Commercial banks can deposit funds at ECB at marginal deposit rate.
- + Commercial banks can borrow funds from ECB at marginal lending rate.

Standing facilities are always available, though typically are used only rarely.



- + The ECB can either expand funds eventually available for lending by households and firms (through provision of additional loans to commercial banks) or contract them (through not renewing (some of) the short-term loans to commercial banks).
- + The ECB in normal times minimizes its lending risk by demanding from commercial banks full collateral for loans in the form of high-quality assets.
- + Empirically, the marginal deposit rate is the floor for the interbank rate of interest (EONIA), and the marginal lending rate is the ceiling for the EONIA.
- + Until late 2008, the EONIA showed at most small deviations from the ECB's main refinancing rate. In this sense, the ECB at least until 2008 seemed to be able to steer the interbank rate of interest. (We postpone a discussion of the global financial crisis that began in 2008 to a later section.)
- + Note that the main refinancing rate is a nominal interest rate.
- + The interest rate, r , that (co-) determines aggregate output demand and appears in the IS -curve, is a real interest rate, though.
- + How are nominal interest rates linked to real interest rates?

Real and Nominal Interest Rates: The Fisher Equation

As the real interest rate r_t captures how many additional units of a good can be consumed in period $t + 1$ when foregoing consumption of one unit of that good in period t , it is linked to the nominal interest rate R_t as follows:

$$\underbrace{r_t}_{\text{real interest rate}} = \underbrace{P_t}_{\text{selling one unit of good in period } t} \cdot \underbrace{(1 + R_t)}_{\text{nominal interest rate (nominal return from } t \text{ to } t + 1)} \cdot \underbrace{\frac{1}{P_{t+1}^e}}_{\text{units of good in } t + 1 \text{ per currency unit, expected in } t} - 1 \quad (57)$$

$$r_t = \frac{1 + R_t}{\frac{P_{t+1}^e}{P_t}} - 1 = \frac{1 + R_t}{1 + \frac{P_{t+1}^e - P_t}{P_t}} - 1 = \frac{1 + R_t}{1 + \underbrace{\frac{P_{t+1}^e - P_t}{P_t}}_{=\pi_t^e}} - 1 \quad (58)$$

$$r_t = \frac{1 + R_t}{1 + \pi_t^e} - 1 = \frac{R_t - \pi_t^e}{1 + \pi_t^e} \approx \underbrace{R_t - \pi_t^e}_{\substack{\text{under } \pi_t^e \\ \text{sufficiently low}}} \quad (59)$$

Equation (59), stating that the real interest rate from period t to $t + 1$ is (approximately) equal to the nominal interest rate from t to $t + 1$ minus the expected inflation rate from t to $t + 1$, is also known as the **Fisher equation**.

Returning to the ECB's main refinancing rate, what explains its variations?

While central banks' decision rationale typically is not tied to a fixed formula, the ECB's and many other central banks' ("normal times") interest rate decisions can be captured surprisingly well through the so-called **Taylor rule**, featuring just four determinants for the nominal monetary policy rate set by a central bank:

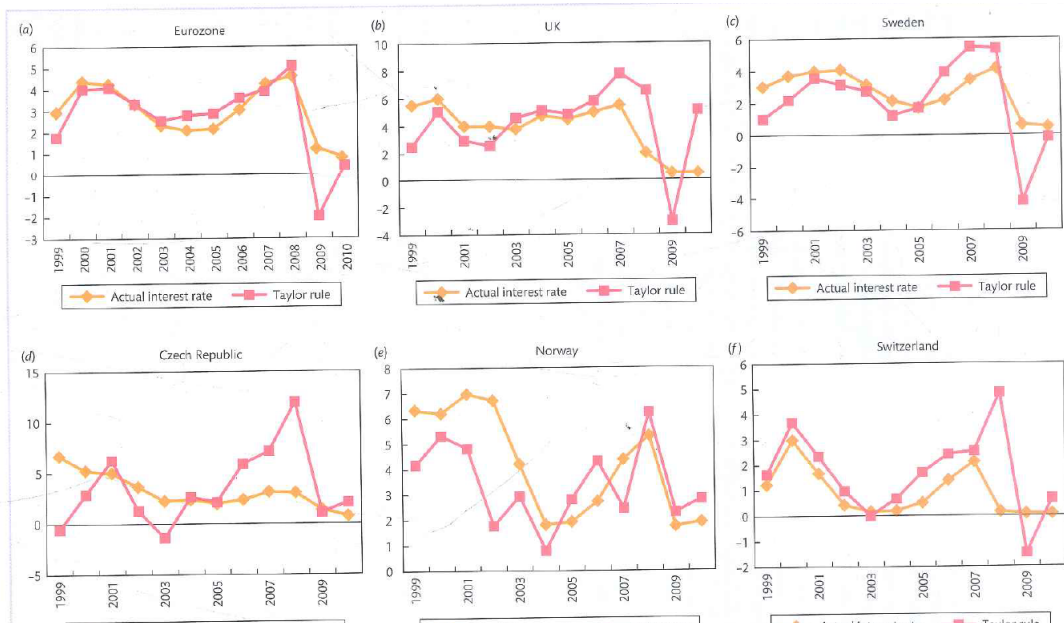
- the "neutral" real monetary policy rate,
- the expected rate of inflation,
- the output gap, and
- deviations of actual inflation from target inflation:

$$R^{MP} = \underbrace{r_n^{MP}}_{\substack{\text{neutral} \\ \text{real rate,} \\ > 0}} + \pi^e + \underbrace{\phi_y}_{>0} \cdot \underbrace{\left(\frac{Y - \bar{Y}}{\bar{Y}} \right)}_{\substack{\text{(relative)} \\ \text{output gap}}} + \underbrace{\phi_\pi}_{>0} \cdot \underbrace{(\pi - \pi^T)}_{\substack{\text{inflation} \\ \text{target}}} \quad (60)$$

The idea underlying the Taylor rule is that the central bank:

- raises the monetary policy rate if macroeconomic activity "overheats" ($Y > \bar{Y}$), or if actual inflation is above target inflation ($\pi > \pi^T$),
- lowers the monetary policy rate if macroeconomic activity is "sluggish" ($Y < \bar{Y}$), or if inflation is below target ($\pi < \pi^T$).

Example of Taylor rules



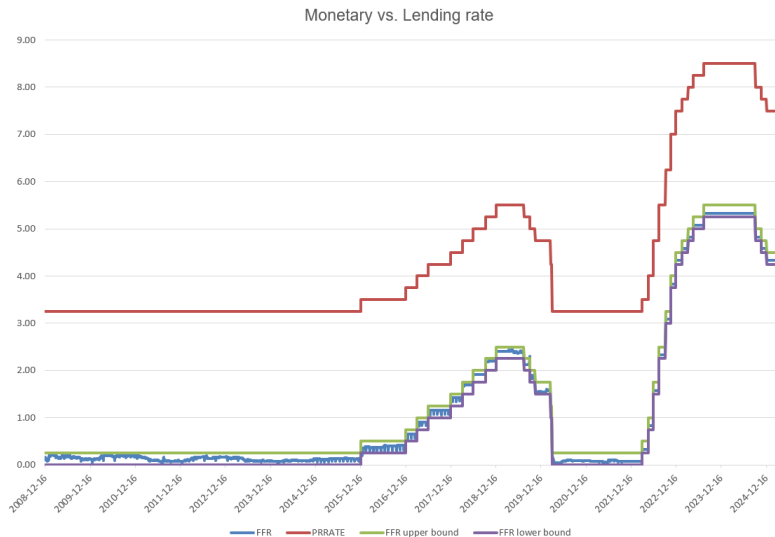
$$\text{If } Y = \bar{Y} \text{ and } \pi = \pi^T, \quad R^{MP}|_{Y=\bar{Y}, \pi=\pi^T} = r_n^{MP} + \pi^e \quad (61)$$

For now, we will treat r_n^{MP} as exogenous and consider its possible determinants later. For the time being, we therefore have $r_n^{MP} = r_0^{MP}$.

Taylor rule in real terms in the short-run:

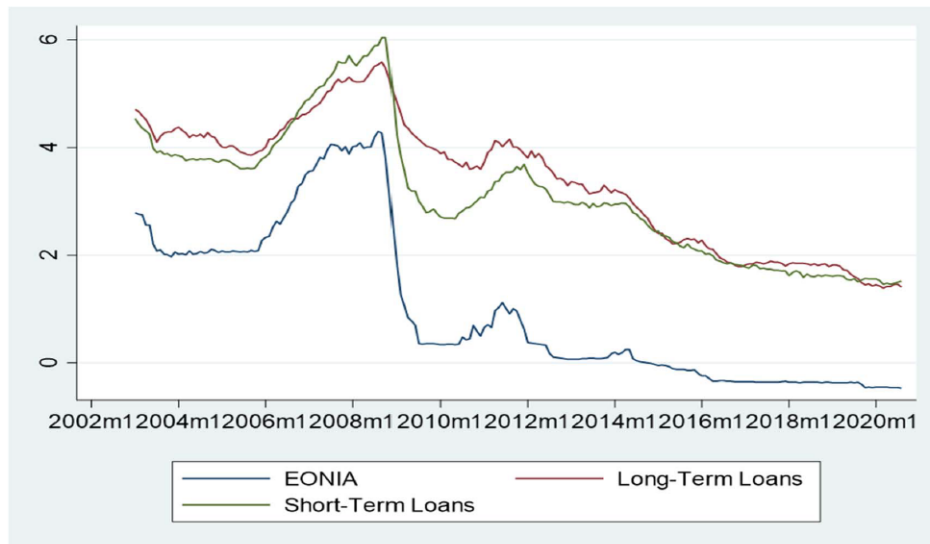
$$\begin{aligned} r^{MP} &\approx R^{MP} - \pi^e = \underbrace{r_n^{MP}}_{=r_0^{MP}} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right) + \underbrace{\phi_\pi}_{=0} \cdot \underbrace{(\pi - \pi^T)}_{=0} \\ \Leftrightarrow \quad r^{MP} &= r_0^{MP} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right) \end{aligned} \quad (62)$$

Do you notice any relationship between the lending rate and the central bank's rate?



Source: Federal Reserve Bank of St. Louis, FRED. Accessed April 2025.

The same pattern holds for the case of the Euro area



Source: The ECB

What drives loan mark-ups? The role of equity

What is driving the mark-ups? In determining how much to lend to a household or firm, a commercial bank considers:

the expected loan return,

the loan risk:

- borrowers may default and the loan collateral (if any) may not provide full cover for all the loan payments a defaulting borrower still does owe,
- given loan risk, even a commercial bank operating under perfect competition (that is a "price taker", leaving determination of r to the credit market) needs to charge a mark-up on its loans,

its ability to bear risk, measured by its equity.

To understand the role of equity, consider the balance sheet of a typical commercial bank:

Assets	Liabilities
Reserves (including cash and reserves at central bank)	Deposits (including customers' checking and saving accounts)
Securities (including bonds and asset-backed securities)	Borrowings (including funds borrowed from central bank and from other commercial banks)
Loans (including industrial and consumer loans)	Other Liabilities
Other Assets	Capital (Equity)

For a given amount of equity, the larger the volume of loans granted, the higher the bank's **leverage** (i.e., the larger the extent to which the bank's loans are funded by debt rather than equity).

Debt, being tax-deductible, often offers a less expensive funding source than equity.

Too much leverage can be perilous:

- In times of distress, debt servicing obligations may exceed the bank's cash flow from loans.
- If the bank becomes insolvent, equity is wiped out and the bank fails.

Regulatory constraints (e.g., minimum capital requirements or leverage caps) limit excessive risk-taking.

We model the commercial bank as choosing its loans-to-equity ratio to maximize expected net return, accounting for loan risk, in mean-variance form:

$$\max_{Lo^S} \left\{ r \cdot \frac{Lo^S}{Eq} - r^{MP} \cdot \frac{Lo^S - Eq}{Eq} - \frac{1}{2} \cdot \underbrace{\chi}_{\text{bank risk aversion}} \cdot \left(\frac{Lo^S}{Eq} \right)^2 \cdot \underbrace{\sigma_{LR}^2}_{\text{loan return risk}} \right\} \quad (64)$$

where Lo^S = loans supplied, Eq = equity. We assume all non-equity funding is secured at rate r^{MP} , and loans to households/firms are not differentiated.

Note that (64) is an example of a mean-variance preference function. Such a preference function holds, for example, for a financial investor with negative exponential utility and normally distributed wealth:

$$u(W) = -\exp(-\underbrace{\lambda}_{\text{coefficient of absolute risk aversion}} \cdot W), \quad \lambda > 0, \quad W \sim \mathcal{N}(\mu^W, (\sigma^2)^W) \quad (65)$$

Under (65), it holds that (proof omitted):

$$\underbrace{E[u(W)]}_{\text{expected utility}} = -\exp\left[-\lambda \cdot \left(\mu^W - \frac{\lambda}{2} (\sigma^2)^W\right)\right] \quad (66)$$

So maximizing expected utility is equivalent to maximizing:

$$\mu^W - \frac{\lambda}{2} (\sigma^2)^W$$

We obtain the following first-order condition from the objective in (64):

$$\frac{r - r^{MP}}{Eq} - \chi \cdot \sigma_{LR}^2 \cdot \frac{Lo^S}{Eq^2} = 0 \quad (67)$$

Solving (67) for Lo^S , we obtain the commercial bank's loan supply:

$$Lo^S = \frac{Eq}{\underbrace{\chi \cdot \sigma_{LR}^2}_{=1/\nu}} \cdot (r - r^{MP}) \quad (68)$$

To obtain the real interest rate, r , that clears the credit market, it remains to combine loan supply with loan demand.

Recalling that we modelled both consumption and investment expenditure as a negative function of the real interest rate, r , we model loan demand as

$$Lo^D = \underbrace{A^{Lo}}_{\substack{\text{autonomous} \\ \text{component} \\ \text{of loan demand}}} - \underbrace{L_{Or}}_{>0} \cdot r, \quad (69)$$

with the parameter L_{Or} denoting the interest-rate sensitivity of loan demand.

Notes:

- (i) The exogenous determinants of loan demand in A^{Lo} include, for example, loan demand from abroad.
- (ii) We do not model loan demand as a function of income, though we had modelled consumption expenditure to be income-sensitive. As their incomes rise, households
 - may take on more debt without increasing their debt-to-income ratios, but also
 - may want to use the additional income to pay down debt, so as to be able to take on more debt when their incomes fall (and thus smooth consumption).

The overall effect of income changes on loan demand is therefore ambiguous, and we do not model it.

To obtain the market-clearing real interest rate, r , at which the commercial banks lend to households and firms, it remains to equate loan supply to loan demand. Setting $Lo^S = Lo^D$, from Equations (68) and (69) we obtain

$$\frac{1}{\nu} \cdot (r - r^{MP}) = A^{Lo} - L_{Or} \cdot r, \quad (70)$$

which in turn yields

$$r = \frac{r^{MP} + \nu \cdot A^{Lo}}{1 + \nu \cdot L_{Or}}, \quad (71)$$

with ν from Equation (68) given by:

$$\nu = \frac{\chi \cdot \sigma_{LR}^2}{Eq}. \quad (72)$$

From (71) and (72), the **mark-up**, $r^{MU} = r - r^{MP}$, that the commercial banks charge for their loans to households and firms **is a function of**

the **risk** associated with these loans (measured through σ_{LR}^2),

the degree of **risk aversion** (χ) of the commercial banks, and

the amount of **equity** (Eq) that the commercial banks hold.

Also from (71), additional factors that affect this mark-up include the autonomous component of loan demand (A^{Lo}), and the interest rate sensitivity of loan demand (L_{Or}).

Focusing on the riskiness of loans, how does the component of the mark-up that is due to this risk (henceforth called the **risk premium**, r^{RP}) respond to an increase of σ_{LR}^2 ?

From Equations (71) and (72) we have

$$\begin{aligned} \frac{\partial r}{\partial \sigma_{LR}^2} &= \frac{\partial r}{\partial \nu} \cdot \frac{\partial \nu}{\partial \sigma_{LR}^2} = \left[\frac{A^{Lo}}{1 + \nu \cdot L_{Or}} - \left(\frac{1}{1 + \nu \cdot L_{Or}} \right)^2 \cdot L_{Or} \cdot (r^{MP} + \nu \cdot A^{Lo}) \right] \cdot \frac{\chi}{Eq} \\ &= \left(\frac{1}{1 + \nu \cdot L_{Or}} \right) \cdot \underbrace{\left[A^{Lo} - L_{Or} \cdot \underbrace{\left(\frac{r^{MP} + \nu \cdot A^{Lo}}{1 + \nu \cdot L_{Or}} \right)}_{=r} \right]}_{=Lo^D} \cdot \frac{\chi}{Eq} > 0. \end{aligned} \quad (73)$$

An increase in the riskiness of loans therefore leads to a higher mark-up, and, as anticipated in Equation (63), we have linked the mark-ups that the commercial banks charge to a risk premium and other factors, such as the determinants of commercial bank equity and of the autonomous component of loan demand:

$$r^{MU} = r^{MU}(\underbrace{r^{RP}}_{(+)}, \text{other factors}).$$

Based on (62) and (63), as the baseline case for the determination of the real interest rate in the financial sector, we consider the financial sector **real interest rate function**

$$r = r(\underbrace{r^{MP}}_{(+)}, \underbrace{r^{RP}}_{(+)}, \text{other factors}), \quad (74)$$

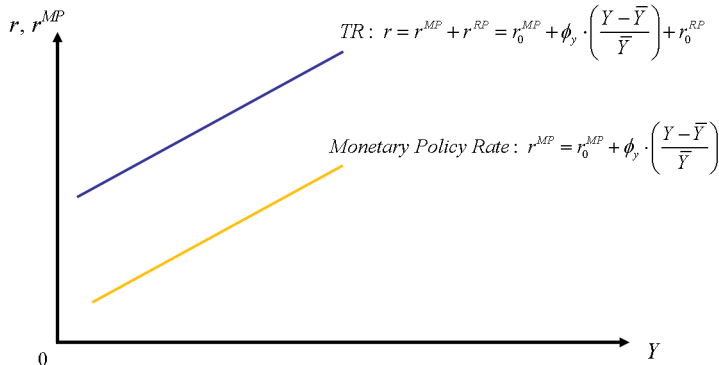
explicitly accounting for the monetary policy rate and the risk premium, and relegating variables other than the risk premium that affect the mark-up between the real interest rate and the monetary policy rate to the "other factors" in Equation (74).

The Short-run TR Curve

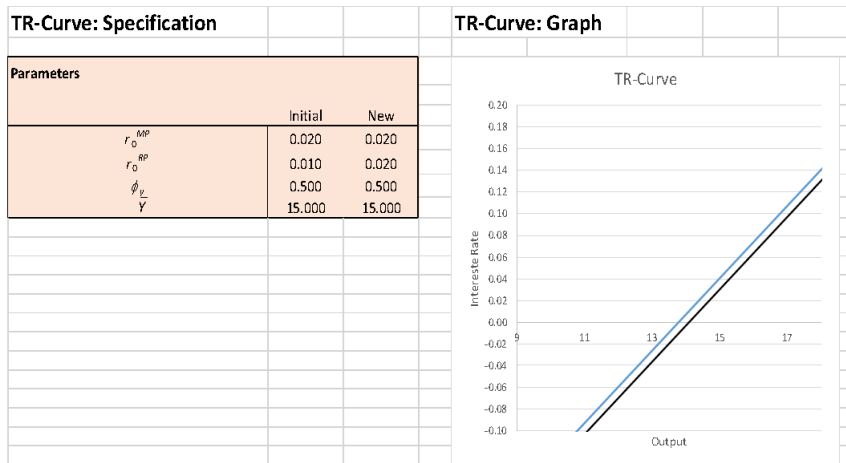
For the short-run quantitative analysis of business cycles, we will typically specialize the financial sector real interest rate function to

$$r = r^{MP} + r^{RP} = r_0^{MP} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right) + r_0^{RP} \quad (75)$$

with the parameters r_0^{MP} , ϕ_y , and r_0^{RP} (that is, for the time being, we will treat the risk premium as exogenous). The relationship in (75) is called the **TR-curve**.



The workbook *TR-Curve.xlsm* allows to study all the determinants of the position and slope of the *TR*-curve.



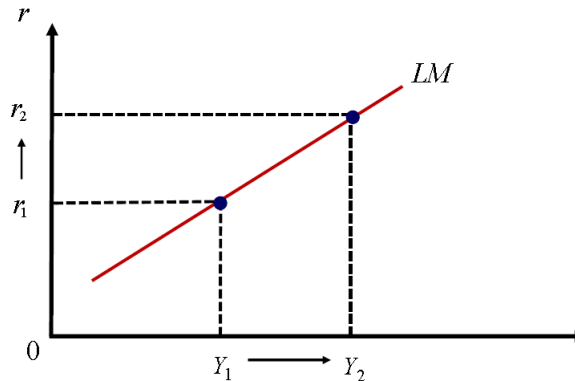
A note on modelling and terminology:

Historically, financial sector outcomes in models of the macroeconomy in the short run have been captured through the so-called **LM-curve**. The LM-curve posits that

- + as the demand for money increases (decreases) with increases (decreases) of income,
- + central banks, aiming to maintain a money supply target, raise the interest rate,
- + as money demand then drops back down to the level of money supply: as the interest rate raises, so does the opportunity cost of holding money rather than holding interest-bearing assets, causing a decline of the demand for money.

The LM-curve thus

- does not distinguish between the monetary policy rate and the interest rate at which households and firms can borrow, and
- depicts central banks that rather than targeting inflation and output (as is modelled by the Taylor rule), target the level of money supply.



The standard in modern macroeconomic analysis has been shifting to carrying out TR-curve based analyses, reflecting observed current central bank behavior as well as the role of financial sector institutions such as commercial banks.

We are now in a position to consider the determination of short-run macroeconomic outcomes as implied by the decision making in the real and financial sectors.

- + We will obtain (short-run) output, Y , and the interest rate, r , by combining the TR -curve (describing the relationship between r and Y in the financial sector) with the IS -curve (describing the relationship between Y and r in the real sector).
- + Once we have determined output and the interest rate, we can then also determine the other key macroeconomic aggregates, including consumption, investment, government expenditure, the trade balance and the exchange rate.

The baseline ***IS-TR model*** consists of the specifications of desired aggregate demand that entered the IS -curve, as well as the financial sector real interest rate function underlying the TR -curve:

- + consumption, from Equation (3):

$$C = C_0 + C_y \cdot (Y - T) - C_r \cdot r, \quad (3)$$

- + investment, from Equation (15):

$$I = I_0 - I_r \cdot r, \quad (15)$$

- + government expenditure, from Equation (21):

$$G = G_0 + G_y \cdot (\bar{Y} - Y), \quad (21)$$

- + the trade balance, from Equations (24) and (25):

$$TB = TB_0 + TB_{ex} \cdot (Y^* - T^*) - TB_{im} \cdot (Y - T) + TB_{\varepsilon} \cdot [\varepsilon_0 - \varepsilon_r \cdot (r - r^*)], \quad (24, 25)$$

Together we can derive the *IS*-curve in Equation (51):

$$Y = \frac{A}{1 - (C_y - TB_{im} - G_y)} - \frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot r \quad (51)$$

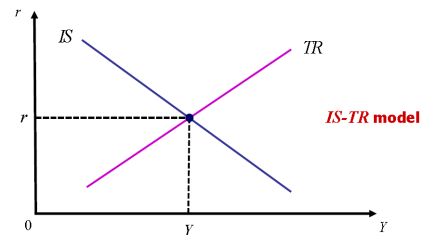
with, from Equation (49):

$$A = C_0 + I_0 + G_0 + TB_0 + TB_\varepsilon \cdot \varepsilon_0 - (C_y - TB_{im}) \cdot T + G_y \cdot \bar{Y} + TB_{ex} \cdot (Y^* - T^*) + TB_\varepsilon \cdot \varepsilon_r \cdot r^* \quad (49)$$

+ the real interest rate, from the *TR*-curve in Equation (75):

$$r = r_0^{MP} + \phi_y \cdot \underbrace{\frac{Y - \bar{Y}}{\bar{Y}}}_{\text{output gap}} + r_0^{RP} \quad (75)$$

The variables that we will continue to keep as exogenous, that is, are determined outside the model, are: T , $\bar{Y}$, Y^* , T^* , and r^* .



The short-run macroeconomic outcome occurs at the intersection of the *IS* curve and *TR* curve: it is the one combination of output and real interest rate that is compatible with the short-run outcomes in both the real and the financial sector.

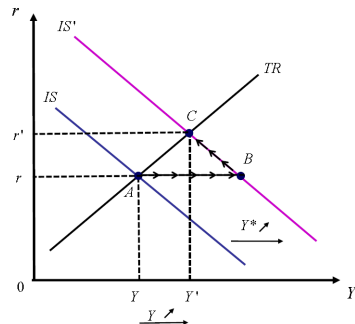
To appreciate better how the *IS-TR* model explains the functioning of the macroeconomy in the short run, let us consider what happens following an increase in foreign income, Y^* . Note at the outset, though, that the *IS-TR* model

is a static model (all variables appearing in the model are dated the same time period), and thus the model

- + pins down the overall changes in the outcomes that occur following any exogenous change, such as an increase in foreign income,
- + but is not specific about the adjustment dynamics, such as in which sequence the various variables adjust.

In what follows, we graph and then discuss the economic rationale of one specific adjustment path, a path that entails a useful separation of the overall changes into a sequence of effects.

- + Starting from the original outcome at Point A, the increase in Y^* causes the *IS*-curve to shift to the right.
 - At the initially market-clearing interest rate, r , output first rises due to the increase in desired aggregate demand (the increase in Y^* implies



- Output also rises because of the Keynesian multiplier effect (chain of effects running from higher output to higher desired aggregate demand to higher output etc.). This implies that the rightward shift of the *IS*-curve is larger than the initial increase in output, that is, is larger than $TB_{ex} \cdot \Delta Y^*$: From the *IS*-curve, we know that for a given interest rate the increase in output taking into account the Keynesian multiplier effect is equal to

$$\frac{TB_{ex} \cdot \Delta Y^*}{1 - \text{MPC}}$$

- + At Point *B*, we are off the *TR*-curve, however: The financial sector needs to adjust.
 - As output has increased from Point *A* to Point *B*, the interest rate according to the central bank's Taylor rule now is too low (reflecting future risks to price stability due to overheating of macroeconomic activity).
 - Simultaneous clearing of the real and financial sectors is restored as we move from Point *B* to Point *C*: The central bank raises the interest rate to a higher level that is in line with the Taylor rule.
 - As the interest rate increases, the movement from Point *B* to Point *C* also involves a partial reduction in output:
 - households decrease consumption expenditure, reducing desired aggregate demand (**first crowding-out effect: consumption channel**),
 - firms decrease investment in physical capital, also reducing desired aggregate demand (**second crowding out effect: investment channel**),
 - the domestic currency's current value appreciates (per the interest parity implication), implying that imports increase and exports decrease, that is, the trade balance decreases, further reducing desired aggregate demand (**third crowding out effect: exchange rate channel**).
- + The new short-run outcome simultaneously clearing the real and financial sectors occurs at Point *C* – involving a higher level of output, a higher domestic interest rate and an appreciated current value of the domestic currency.

Solving the *IS-TR* Model

A key virtue of having algebraic equations for the *IS*- and *TR*-curves is that we can calculate the magnitudes of the changes in output and the interest rate (as well as in consumption, investment, government expenditure, the trade balance and the exchange rate). Re-writing the *TR*-curve, (75),

$$r = r_0^{MP} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right) + r_0^{RP},$$

as

$$Y = \bar{Y} + \left(\frac{\bar{Y}}{\phi_y} \right) \cdot (r - r_0^{MP} - r_0^{RP}), \quad (76)$$

to obtain the interest rate in the short-run macroeconomic outcome, we equate output as predicted by the *TR*-curve, (76), to output as predicted by the *IS*-curve, (51):

$$Y = \underbrace{\frac{A - (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot r}{1 - (C_y - TB_{im} - G_y)}}_{\text{from IS-curve (51)}} = \underbrace{\bar{Y} + \left(\frac{\bar{Y}}{\phi_y} \right) \cdot (r - r_0^{MP} - r_0^{RP})}_{\text{from TR-curve re-written as (76)}}. \quad (77)$$

Note that (77) is one equation in one unknown, namely r . We can thus use Equation (77) to solve for r :

$$\frac{A}{1 - (C_y - TB_{im} - G_y)} - \bar{Y} \cdot \left(1 - \frac{r_0^{MP} + r_0^{RP}}{\phi_y} \right) = \left(\frac{\bar{Y}}{\phi_y} + \frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \right) \cdot r$$

$$\Leftrightarrow r = \frac{\frac{A}{1-(C_y-TB_{im}-G_y)} - \bar{Y} \cdot \left(1 - \frac{r_0^{MP} + r_0^{RP}}{\phi_y}\right)}{\frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1-(C_y-TB_{im}-G_y)} + \frac{\bar{Y}}{\phi_y}}. \quad (78)$$

Equation (78) gives us the *IS-TR* model-implied **interest rate** in the short-run macroeconomic outcome.

Having obtained the rate of interest, we can calculate the *IS-TR* model-implied level of **output** in the short-run macroeconomic outcome from the *IS*-curve, Equation (51),

$$Y = \frac{A}{1-(C_y-TB_{im}-G_y)} - \frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1-(C_y-TB_{im}-G_y)} \cdot r,$$

substituting for r from Equation (78).

We are then also in a position to plug back into our consumption, investment, government expenditure, trade balance and exchange rate functions to obtain the short-run outcomes for C , I , G , TB and ε .

Suppose $A = 9.78$, $C_y = 0.6$, $G_y = 0.15$, $TB_{im} = 0.05$, $TB_{ex} = 0.01$,
 $C_r = 4$, $I_r = 16$, $\bar{Y} = 15$, $r_0^{MP} = 0.02$, $r_0^{RP} = 0.01$, $\phi_y = 0.5$, $TB_\varepsilon = 1$, $\varepsilon_r = 6$.

Using these parameter values in Equation (78), we obtain $r = 0.03$. Making use of this value of r and those of the parameters above in the IS -curve, Equation (51), we obtain $Y = 15$.

Now suppose that Y^* increases by 15: $\Delta Y^* = 15$. As from Equation (49) we have that

$$A = C_0 + I_0 + G_0 + TB_0 + TB_\varepsilon \cdot \varepsilon_0 - (C_y - TB_{im}) \cdot T + C_y \cdot \bar{Y} + TB_{ex} \cdot (Y^* - T^*) + TB_\varepsilon \cdot \varepsilon_r \cdot r^*,$$

this increase in foreign income implies a change in the autonomous component of aggregate demand by

$$\Delta A = \underbrace{\frac{\partial A}{\partial Y^*}}_{=TB_{ex}} \cdot \Delta Y^* = TB_{ex} \cdot \Delta Y^* = 0.15. \quad (79)$$

From Equation (78), we then have that

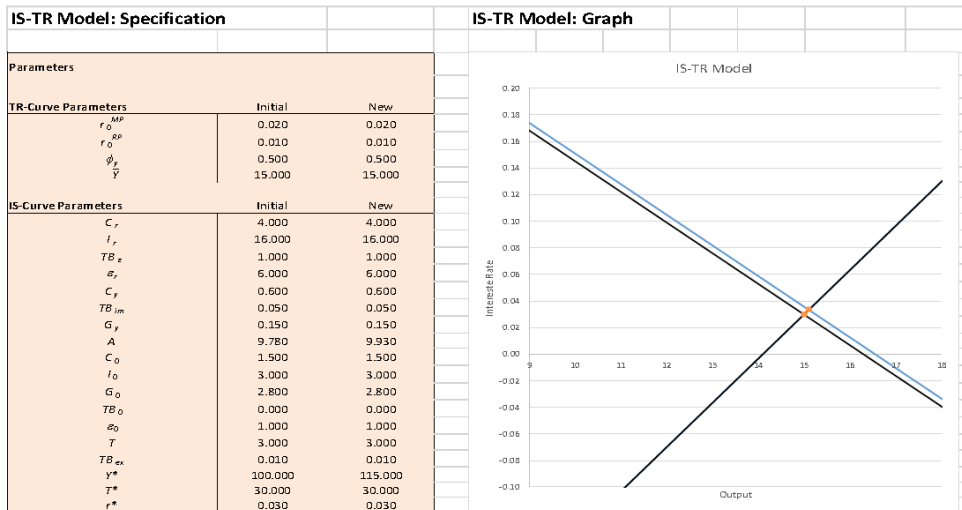
$$\Delta r = \frac{\partial r}{\partial A} \cdot \Delta A = \frac{1}{1 - (C_y - TB_{im} - G_y)} \cdot \frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y) + \frac{\bar{Y}}{\phi_y}} \cdot \Delta A \approx 0.003, \quad (80)$$

and from the IS -curve, Equation (51), and Equation (80) that

$$\begin{aligned} \Delta Y &= \underbrace{\frac{\partial Y}{\partial A} \cdot \Delta A}_{\text{direct effect}} + \underbrace{\frac{\partial Y}{\partial r} \cdot \frac{\partial r}{\partial A} \cdot \Delta A}_{\text{indirect effect}} = \frac{\partial Y}{\partial A} \cdot \Delta A + \frac{\partial Y}{\partial r} \cdot \underbrace{\Delta r}_{=\frac{\partial r}{\partial A} \cdot \Delta A} \\ &= \frac{1}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta A - \frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r \approx 0.102. \end{aligned} \quad (81)$$

Note that Equation (81) also gives us a useful decomposition of the overall change in output.

The workbook IS-TR-Model.xlsm allows to study how changes in the autonomous components of aggregate demand and/or other model parameters affect output in the short-run macroeconomic outcome, ...



... how these output effects may be decomposed, and how the other endogenous variables change:

IS-TR Model: Keynesian Multiplier and Consumption, Investment as well as Exchange Rate Channels: Quantitative Results		Initial and New Short-Run Macroeconomic Outcomes				
Transmission Channels for Changes in Output Following Changes Exclusively in the Components of Autonomous Demand		Initial and New Short-Run Macroeconomic Outcomes Following Changes in Any of the Model Parameters				
Keynesian Multiplier			Initial	New	Absolute Change	Percentage Change
Stimulus: Change in A	0.150	Output	15.000	15.102	0.102	0.682
Size of Multiplier	1.667	Interest Rate	0.030	0.033	0.003	11.364
Change in Output	0.250	Monetary Policy Rate	0.020	0.023	0.003	17.045
Interest Rate Change: Consumption Channel		Consumption	8.580	8.628	0.048	0.556
		Investment	2.520	2.465	-0.055	-2.165
Change in Output	-0.023	Government Expenditure	2.800	2.785	-0.015	-0.548
Interest Rate Change: Investment Channel		Trade Balance	1.100	1.224	0.124	11.312
		Exchange Rate	1.000	0.980	-0.020	-2.045
Change in Output	-0.091					
Interest Rate Change: Exchange Rate Channel						
Change in Output	-0.034					
Total Change in Output						
Sum of Effects	0.102					

How do changes in I_r affect business cycle outcomes?

Recalling the IS -curve (51), re-written in the form of Equation (52),

As we had seen in the data, not all business cycles are alike. Business cycles may vary

- + across countries or,
- + across time periods within a given country.

One reason is that the **shocks (unanticipated changes)** to the autonomous components of aggregate demand that are the starting point for a business cycle fluctuation may vary in size across countries and/or time periods.

Another reason is that the **propagation** of the shocks may vary in size across countries and/or time periods.

The strength of the shock propagation can vary with the magnitude of the parameters entering the IS - TR model. To appreciate this point, consider, say, the sensitivity of investment to the interest rate, I_r :

How do changes in I_r affect business cycle outcomes? Recalling the TR -curve (75),

$$r = \underbrace{r_0^{MP} + r_0^{RP}}_{=TR_0} - \underbrace{\frac{\phi_Y}{Y}}_{=TR_1} \cdot Y,$$

both the intercept of the TR -curve, TR_0 , and the slope of the TR -curve, TR_1 , are insensitive to I_r , as

$$\frac{\partial TR_0}{\partial I_r} = 0, \quad (82)$$

$$\frac{\partial TR_1}{\partial I_r} = 0. \quad (83)$$

Changes in the sensitivity of investment to the interest rate, I_r , may therefore affect business

Recalling the IS -curve (51), re-written in the form of Equation (52),

$$r = \underbrace{\frac{A}{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}}_{=IS_0^r} - \underbrace{\frac{1 - (C_y - TB_{im} - G_y)}{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}}_{=IS_1^r} \cdot Y,$$

note that the sensitivity of the intercept of the graphed IS -curve, IS_0^r , to I_r is given by

$$\frac{\partial IS_0^r}{\partial I_r} = \underbrace{\frac{\partial A}{\partial I_r}}_{=0} \cdot \frac{1}{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r} + \frac{\partial [C_r + I_r + TB_\varepsilon \cdot \varepsilon_r]^{-1}}{\partial I_r} \cdot A < 0. \quad (84)$$

Also, the sensitivity of the slope of the graphed IS -curve, $-IS_1^r$, to I_r is given by

$$\frac{\partial (-IS_1^r)}{\partial I_r} = \frac{\partial}{\partial I_r} \left[\frac{1 - (C_y - TB_{im} - G_y)}{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r} \right] = \frac{1 - (C_y - TB_{im} - G_y)}{(C_r + I_r + TB_\varepsilon \cdot \varepsilon_r)^2} > 0. \quad (85)$$

Thus, the larger I_r , the stronger the investment channel: changes in r that occur as the central bank is raising the monetary policy rate (in expansions) or is lowering the monetary policy rate (in recessions) cause larger crowding out/in effects.

Consider a numerical example: Instead of $I_r = 16$, suppose that we have $\tilde{I}_r = 34$. How does this affect short-run macroeconomic outcomes?

To obtain the initial short-run interest rate prevailing under $\tilde{l}_r = 34$, simply replace l_r in Equation (78) by $\tilde{l}_r$:

$$r = \frac{A}{1 - (C_y - TB_{im} - G_y)} - \bar{Y} \cdot \left(1 - \frac{r_0^{MP} + r_0^{RP}}{\phi_y}\right) \cdot \frac{C_r + \tilde{l}_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y) + \frac{\bar{Y}}{\phi_y}} \approx 0.021. \quad (86)$$

Making use of this value of r in the IS -curve, Equation (51), we obtain $Y \approx 14.739$. Following an increase of foreign output, Y^* , again by 15, we now have from Equation (80) (again with l_r replaced by $\tilde{l}_r$) that

$$\Delta r = \frac{\partial r}{\partial A} \cdot \Delta A = \frac{\frac{1}{1 - (C_y - TB_{im} - G_y)}}{\frac{C_r + \tilde{l}_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} + \frac{\bar{Y}}{\phi_y}} \cdot \Delta A \approx 0.002, \quad (87)$$

and from Equation (81) (once more with l_r replaced by $\tilde{l}_r$) and Equation (87) that

$$\Delta Y = \frac{1}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta A - \frac{C_r + \tilde{l}_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r \approx 0.073. \quad (88)$$

Under the increase of the sensitivity of investment to the interest rate considered in this numerical example, the change in output is smaller than before, that is, the short-run expansion is weaker. Decomposing the overall change in output:

+ Change in output due to Keynesian Multiplier effect:

$$\frac{\Delta A}{1 - (C_y - TB_{im} - G_y)} = 0.25,$$

independent of the sensitivity of investment to the interest rate.

- + Change in output due to crowding out of consumption (consumption channel):

$$-\frac{C_r}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r \approx -0.016,$$

- reflecting that under increased sensitivity of investment to the interest rate the crowding out effects due to the consumption channel are smaller, as the central bank increases the monetary policy rate by less.

- + Change in output due to crowding out of investment (investment channel):

$$-\frac{I_r}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r \approx -0.137,$$

reflecting that under increased sensitivity of investment to the interest rate the crowding out effects due to the investment channel are larger.

- + Change in output due to crowding out of the trade balance (exchange rate channel):

$$-\frac{TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r \approx -0.024.$$

reflecting that under increased sensitivity of investment to the interest rate the crowding out effects due to the exchange rate channel are smaller, as the central bank increases the monetary policy rate by less.

- + The reason as to why in our numerical example the short-run expansion following the increase of the sensitivity of investment to the interest rate is weaker than before is that the larger crowding out effects under the investment channel dominate the weaker crowding out effects under the consumption and exchange rate channels.
- + As one further illustration of the effects of changes in model parameters on business cycle outcomes, let us examine the parameter measuring the sensitivity of the central bank's setting of the monetary policy rate to the level of output: ϕ_y .

How do changes in ϕ_y affect business cycle outcomes?

Recalling the IS -curve (51), re-written in the form of Equation (52),

$$r = \underbrace{\frac{A}{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}}_{=IS_0^r} - \underbrace{\frac{1 - (C_y - TB_{im} - G_y)}{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}}_{=IS_1^r} \cdot Y,$$

both the intercept of the graphed IS -curve, IS_0^r , and the slope of the graphed IS -curve, $-IS_1^r$, are insensitive to ϕ_y , as

$$\frac{\partial IS_0^r}{\partial \phi_y} = 0, \quad (89)$$

and

$$\frac{\partial(-IS_1^r)}{\partial \phi_y} = 0. \quad (90)$$

Changes in the sensitivity of the central bank's setting of the monetary policy rate to the level of output may therefore affect the business cycle only through the TR -curve.

Recalling the TR -curve (75),

$$r = \underbrace{r_0^{MP} + r_0^{RP}}_{=TR_0} + \underbrace{\frac{\phi_y}{Y}}_{=TR_1} \cdot Y,$$

Note that the sensitivity of the intercept of the TR -curve, TR_0 , to ϕ_y is given by

$$\frac{\partial TR_0}{\partial \phi_y} = -1 < 0, \quad (91)$$

and that the sensitivity of the slope of the TR -curve, TR_1 , to ϕ_y is given by

$$\frac{\partial TR_1}{\partial \phi_y} = \frac{1}{\bar{Y}} > 0. \quad (92)$$

Thus, the larger ϕ_y , the steeper the TR -curve: changes in output that occur in response to shocks to the autonomous components of aggregate demand are met by the central bank with larger increases of the monetary policy rate (in expansions) or decreases of the monetary policy rate (in recessions), causing larger crowding out/in effects.

Consider a numerical example: Instead of $\phi_y = 0.5$, suppose that we have that $\tilde{\phi}_y = 2$. How does this affect short-run macroeconomic outcomes?

To obtain the initial short-run interest rate prevailing under $\tilde{\phi}_y = 2$, simply replace ϕ_y in Equation (78) by $\tilde{\phi}_y$:

$$\begin{aligned} r &= \frac{A}{1 - (C_y - TB_{im} - G_y)} - \bar{Y} \cdot \left(1 - \frac{r_0^{MP} + r_0^{RP}}{\tilde{\phi}_y} \right) \cdot \frac{C_r + \tilde{I}_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y) + \frac{\bar{Y}}{\tilde{\phi}_y}} \\ &= 0.03 \end{aligned} \quad (93)$$

Making use of this value of r in the IS -curve, Equation (51), we obtain $Y = 15$.

Following an increase of foreign output, Y^* , again by 15, we now have from Equation (80) (**again with I_r replaced by $\tilde{I}_r$**) that:

$$\Delta r = \frac{\partial r}{\partial A} \cdot \Delta A = \frac{\frac{1}{1 - (C_y - TB_{im} - G_y)}}{\frac{C_r + \tilde{I}_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} + \frac{\bar{Y}}{\phi_y}} \cdot \Delta A \approx 0.005 \quad (94)$$

From Equations (81) and (94):

$$\Delta Y = \frac{1}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta A - \frac{C_r + \tilde{I}_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r \approx 0.037$$

Under the increase of the sensitivity of the central bank's setting of the monetary policy rate to the level of output considered in this numerical example, the change in output is smaller than before, that is, the short-run expansion is weaker.

Decomposing the overall change in output:

+ Change in output due to Keynesian Multiplier effect:

$$\frac{\Delta A}{1 - (C_y - TB_{im} - G_y)} = 0.25$$

+ Change in output due to crowding out of consumption (consumption channel):

$$- \frac{C_r}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r \approx -0.033$$

- + Change in output due to crowding out of investment (investment channel):

$$-\frac{I_r}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r \approx -0.131$$

- + Change in output due to crowding out of trade balance (exchange rate channel):

$$-\frac{TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r \approx -0.049$$

The reason as to why in our numerical example the short-run expansion following the increase of the sensitivity of the central bank's setting of the monetary policy rate to the level of output is weaker than before is that the central bank adjusts the monetary policy rate more strongly to any changes in output, causing uniformly larger crowding out effects.